

Web3 & Digital Assets

Blockchain basics, smart contracts, token economics, and the future of digital ownership

- Level: All levels
- Format: Self-paced micro-lessons + exercises
- Estimated time: 6-10 hours
- Last update: 2025-12-26

Web3 introduces programmable ownership: digital assets, smart contracts, and decentralized networks. This module teaches fundamentals, key risks, and practical frameworks for evaluating digital assets responsibly.

Rule: Understand the technology and the risk. Never invest in what you cannot explain in one page.

1. Learning Outcomes

By the end of this module, the learner will be able to:

- Explain how blockchains, wallets, and smart contracts work at a high level
- Understand tokens, NFTs, and token economics basics
- Identify common risks: custody, smart contract risk, scams, volatility
- Create a framework to evaluate a digital asset project

Prerequisites

- Basic investing knowledge
- Ability to follow security best practices

2. Curriculum Map (Suggested Path)

Unit	Focus	Key Deliverable
1	Blockchain and wallets fundamentals	Web3 basics sheet
2	Smart contracts and DeFi concepts	DeFi risk notes
3	Token economics and project evaluation	Evaluation framework
4	Security, custody, and scams prevention	Security checklist
5	Portfolio approach and risk controls	Digital asset policy

3. Core Concepts

Blockchain basics

A blockchain is a shared ledger. It enables verification without a central authority. Different chains trade off speed, cost, and decentralization.

Smart contracts and DeFi

Smart contracts are code that executes rules. DeFi uses smart contracts for lending, trading, and yield strategies - with unique risks.

Token economics

Tokens are not all the same. Supply schedules, utility, governance, and demand drivers matter.

- Total supply vs circulating supply
- Inflation/emissions and unlock schedules
- Utility and value capture
- Governance rights

Security and custody

Most losses happen from poor security, not market moves. Use layered security and verify every transaction.

4. Practical Toolkit

Use these templates and checklists while you learn. Keep them in your investor journal.

Web3 Project Evaluation (One Page)

Avoid hype-driven decisions.

- What problem does it solve?
- Who are the builders and what is their track record?
- Token utility and value capture
- Supply schedule and unlock risks
- Security audits and risk history
- Catalysts and adoption metrics

Security Checklist

Protect your digital assets.

- Use hardware wallet for long-term storage
- Enable 2FA and strong passwords
- Never share seed phrase
- Verify URLs and contracts
- Send small test transactions first

Digital Asset Policy

Keep exposure within your risk budget.

- Define maximum allocation (e.g., 0-10%)
- Use position sizing and stop rules
- Avoid leverage as a default
- Document every investment thesis

5. Hands-on Exercises

Explain Web3 in 10 sentences

If you cannot explain it simply, you do not understand it yet.

- Write 10 sentences: blockchain, wallet, smart contract, token
- Define 3 risks
- Define 2 safe practices

Output: One-page explanation

Evaluate one project

Use the One-Page Evaluation.

- Pick one known digital asset project
- Fill evaluation steps
- Write go/no-go decision and why

Output: Evaluation sheet

Create your security setup

Write your security plan (no sensitive details).

- List security layers you will use
- Write a recovery plan approach
- Define rules for transaction verification

Output: Security plan

6. Quick Assessment (Self-Check)

Answer in 5-10 minutes. If you miss more than 2 questions, review Units 1-3.

- 1. What is a blockchain in simple terms?
- 2. What is a smart contract and what risk does it introduce?
- 3. Why does token supply schedule matter?
- 4. Name three common Web3 risks.
- 5. What is custody and why is it important?

7. Mini Glossary

Wallet: Software or hardware that holds keys to control assets.

Private key: Secret key that authorizes transactions.

Smart contract: Code that runs on-chain and executes rules.

DeFi: Decentralized finance built on smart contracts.

NFT: Non-fungible token representing unique ownership.

8. Next Steps

Use Risk Management and Behavioral Psychology to control volatility and avoid emotional decisions in digital assets.

- Adopt a written security checklist.
- Evaluate projects using a one-page framework.
- Keep allocation within a defined policy.